Extra Practice 1: Fractions Refresher: Mixed Numbers

A. Skills

1. Write each improper fraction as a mixed number in simplest form.

(a) $\frac{15}{4}$ (b) $\frac{26}{3}$ (c) $\frac{42}{8}$ (d) $\frac{35}{10}$

2. Write each mixed number as an improper fraction.

(a) $1\frac{7}{8}$ (b) $4\frac{3}{10}$ (c) $9\frac{2}{3}$ (d) $6\frac{5}{12}$

3. Add. Write each answer in simplest form.

(a) $2\frac{3}{10} + 4\frac{5}{10}$ (b) $3\frac{1}{2} + 1\frac{2}{3}$ (c) $7\frac{5}{8} + 3\frac{15}{16}$

4. Subtract. Write each answer in simplest form.

(a) $8\frac{5}{6} - 3\frac{1}{6}$ (b) $4\frac{1}{5} - 1\frac{1}{2}$ (c) $12 - 7\frac{7}{16}$

5. Multiply. Write each answer in simplest form.

(a) $5 \times 3\frac{1}{4}$ (b) $\frac{3}{4} \times 2\frac{2}{3}$ (c) $2\frac{1}{2} \times 3\frac{1}{5}$ (d) $1\frac{1}{3} \times 1\frac{1}{8}$

6. Write (a) and (b) as mixed numbers in simplest form. Write (c) as a decimal.

(a) 3.25 (b) 7.6 (c) $2\frac{5}{16}$

B. Applying

7. A quilter sews three strips of fabric side by side. The strips are $2\frac{1}{4}$ in, $3\frac{5}{8}$ in and $1\frac{1}{2}$ in wide. How wide is the finished piece? (Ignore the seams.)

8. Last September, Maya was $62\frac{3}{4}$ in tall. This September she is $64\frac{1}{8}$ in tall. How much did she grow?

9. A bannock recipe uses $2\frac{2}{3}$ cups of flour. Owen is making $2\frac{1}{2}$ batches for a community supper.

(a) How much flour does he need?

(b) He has 6 cups of flour. How much more does he need?

10. A plumber has a 10 ft length of pipe. She cuts 4 pieces, each $2\frac{3}{8}$ ft long. Ignoring the saw cuts, how much pipe is left?

11. Which is the best estimate of $5\frac{7}{8} \times 3\frac{1}{4}$?

A. 15 **B.** 18 **C.** 24 **D.** 9

12. Numerical Response A stack has 6 sheets of plywood. Each sheet is $\frac{5}{8}$ in thick. How tall is the stack, in inches? Write your answer as a decimal.

C. Stretch

13. Thomas cuts pieces that are each $11\frac{3}{4}$ in long from a 96 in board. Each saw cut turns $\frac{1}{8}$ in of wood into sawdust. How many full pieces can he cut? Show your work.

14. Find the missing number.

(a) $2\frac{3}{4} + \square = 6\frac{1}{8}$

(b) $4 \times \square = 9$ (write it as a mixed number)

Extra Practice 2: The SI System: Units, Prefixes and Referents

A. Skills

1. Complete each statement.

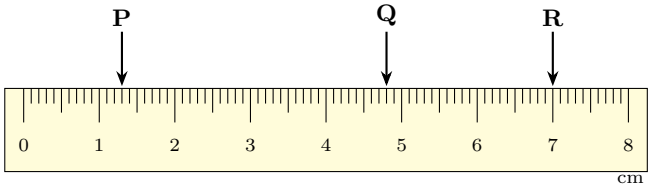
(a) 1 dam = _____ m (b) 1 hm = _____ m (c) 1 dm = _____ m

(d) 1 cL = _____ L (e) 1 mg = _____ g (f) 1 kL = _____ L

2. Choose the most sensible unit: mm, cm, m, km, mg, g, kg, mL or L.

(a) the length of a city bus		(e) the vanilla in a cookie recipe	
(b) the mass of a hockey puck		(f) the drive from Calgary to Lethbridge	
(c) the water in a bathtub		(g) the mass of a moose	
(d) the width of a grain of rice		(h) the vitamin C in one gummy	

3. Read the ruler at P, Q and R. Write each reading in centimetres and in millimetres.



	cm	mm
P		
Q		
R		

4. Change each measurement.

(a) 6.8 m = _____ cm (b) 925 mm = _____ m (c) 0.35 km = _____ m (d) 47 cm = _____ mm

(e) 5600 g = _____ kg (f) 0.08 L = _____ mL (g) 12 cm = _____ m (h) 3.5 g = _____ mg

5. Write $<$, $>$ or $=$ in each circle.

(a) 4.2 cm ◯ 40 mm (b) 0.75 km ◯ 750 m (c) 1.08 m ◯ 110 cm (d) 900 g ◯ 0.85 kg

B. Applying

6. A courier has four parcels: 1.2 kg, 1150 g, 0.98 kg and 1025 g. Put them in order from lightest to heaviest.

7. A nickel has a mass of about 4 g. A bank roll holds 40 nickels.

(a) What is the mass of one roll, in grams?

(b) A bag holds 25 rolls. What is the mass of the bag of nickels, in kilograms?

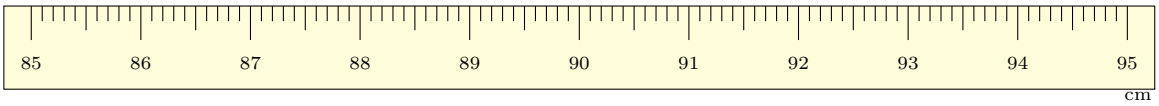
8. A carpenter has a board that is 2.4 m long. She cuts shelves that are each 45 cm long. How many shelves can she cut? How much of the board is left? (Ignore the width of the saw cuts.)

9. Noor is going on a day hike. She wants to carry 3.5 L of water. Her bottles each hold 750 mL. How many bottles does she need to fill?

10. A building plan says a cabinet is 915 mm wide. Ravi's tape measure is numbered in centimetres.

(a) What reading on the tape should Ravi look for?

(b) Draw an arrow on the tape at that reading.



11. The most reasonable capacity for a kitchen sink is

A. 20 mL **B.** 20 L **C.** 2000 L **D.** 0.2 L

12. Numerical Response A 5 km fun run is split into sections that are each 250 m long. The number of sections is _____.

C. Stretch

13. Kai's bike ride to school is 2 km 350 m.

(a) Write the distance in metres, and in kilometres.

(b) He rides to school and back home 5 days a week. How far does he ride in a week, in kilometres?

14. A bottle of plant food says: mix 5 mL of plant food with each 1 L of water.

(a) How much plant food goes in a 4.5 L watering can?

(b) How much plant food goes in a 750 mL spray bottle?

15. Maya says, "To change metres to millimetres, multiply by 100, because milli is two steps below the base unit." Use the staircase to explain her mistake. Then change 1.2 m to millimetres.

Extra Practice 3: The Imperial System and Reading a Tape Measure

A. Skills

1. Fill in the missing number or unit.

(a) _____ in = 1 ft (b) 3 ft = 1 _____ (c) 36 _____ = 1 yd

(d) 5280 _____ = 1 mi (e) 1760 yd = 1 _____

2. Fractions of an inch.

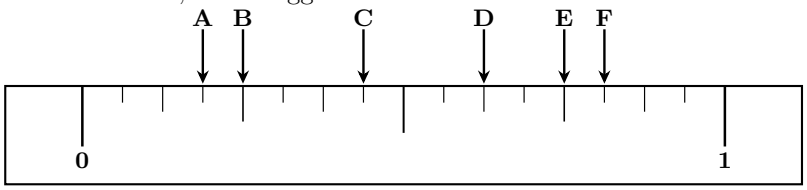
(a) Write in simplest form.

$\frac{10}{16}$ _____ $\frac{4}{16}$ _____ $\frac{14}{16}$ _____ $\frac{6}{16}$ _____

(b) What kind of line is it on a ruler (half, quarter, eighth or sixteenth line)?

$\frac{3}{8}$ _____ $\frac{3}{4}$ _____ $\frac{9}{16}$ _____

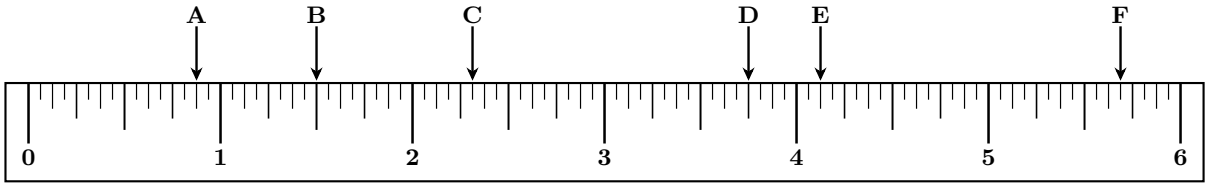
3. This is one inch, made bigger. Write the fraction at each arrow in simplest form.



A _____ B _____ C _____

D _____ E _____ F _____

4. Read the length at each arrow. The ruler is real size.



A _____ B _____ C _____ D _____ E _____ F _____

5. Write each length with symbols or in words.

Write using the symbols $'$ and $''$.

(a) 7 feet 4 inches _____

(b) 5 feet $2\frac{1}{4}$ inches _____

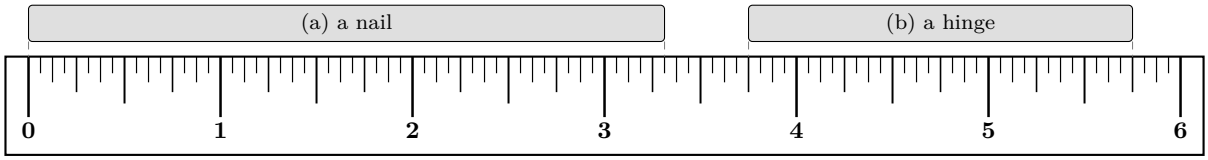
Write in words.

(c) $12'$ _____

(d) $6'\,0\frac{1}{2}''$ _____

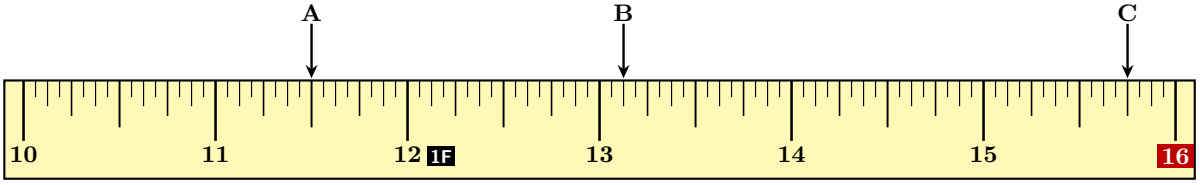
B. Applying

6. How long is each object? The ruler is real size.



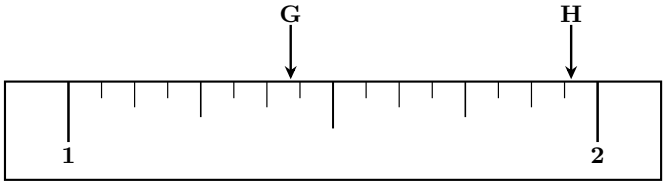
(a) _____ (b) _____

7. Read each arrow on this part of a tape measure (real size). Write it in inches, then in feet and inches if it is 12 in or more.



A _____ B _____ = _____ C _____ = _____

8. This is the part of a ruler from 1 in to 2 in, made bigger. Measure each arrow to the nearest $\frac{1}{16}$ in and to the nearest $\frac{1}{8}$ in.



	nearest $\frac{1}{16}$ in	**nearest $\frac{1}{8}$ in**
G		
H		

9. A carpenter's drill bits are $\frac{5}{16}''$, $\frac{1}{8}''$, $\frac{3}{16}''$, $\frac{1}{4}''$ and $\frac{3}{8}''$.

(a) Put them in order from smallest to largest.

(b) Before putting in a $\frac{1}{4}''$ screw, she drills a hole one size smaller than the screw. Which bit does she use?

10. A garage door opening is $7'\,0''$ high. Lucas's van with a roof rack is $6'\,10\frac{1}{2}''$ high. Will the van fit through the door? By how much?

11. On a ruler marked in sixteenths, an arrow is on the 5th line past the 3. The length is

A. $3\frac{5}{8}$ in **B.** $3\frac{5}{16}$ in **C.** $5\frac{3}{16}$ in **D.** $3\frac{1}{2}$ in

12. Numerical Response How many $\frac{1}{16}$ in spaces are there from the 2 in line to the $3\frac{1}{2}$ in line?

C. Stretch

13. Maya wants to drill a hole in the centre of a shelf that is $5\frac{3}{4}$ in wide. How far from the edge should she mark the centre?

14. Owen hooks his tape on the end of a pipe and marks it at $28\frac{3}{16}$ in and at $35\frac{1}{2}$ in. How long is the piece between the marks? Check by adding.

15. Kira notices that the hook on her new tape measure wiggles. She thinks the tape is broken. Explain why the hook is loose on purpose.

Extra Practice 4: Converting Length Within SI and Within Imperial

A. Skills

1. Convert each length.

(a) 7.3 m to cm

(b) 465 cm to m

(c) 3.9 km to m

(d) 28 mm to cm

(e) 0.4 m to mm

(f) 2 600 000 mm to km

2. Convert each length.

(a) 11 ft to in

(b) 30 in to ft

(c) 7 yd to ft

(d) 4 mi to yd

(e) 108 in to yd

(f) 13 200 ft to mi

3. Convert each length.

(a) $6'\,1''$ to inches

(b) 92 in to ft and in

(c) 50 in to ft and in

(d) $10'\,3''$ to decimal feet

(e) 3.5 ft to ft and in

(f) $1'\,8''$ to decimal feet (nearest hundredth)

4. Find each answer in feet and inches, with fewer than 12 inches.

(a) $2'\,11'' + 4'\,5''$

(b) $10'\,1'' - 3'\,7''$

(c) $3 \times 3'\,5''$

B. Applying

5. A tape measure shows a window is 1.85 m wide. Write the width in cm and in mm. A door is 2032 mm tall. Write its height in m.

6. Each school day Zoe bikes 2.4 km from home to school, then 650 m from school to her job, then 2.9 km from her job back home.

(a) How far does she bike each day, in km?

(b) How far does she bike in a 5-day week?

7. A hallway has two long walls that are each $14' 7''$ and an end wall that is $3' 10''$. How much baseboard trim does it need?

8. A laser measure shows that a room is 11.6 ft long. Write the length in feet and inches, to the nearest inch.

9. A new hockey stick is 60 in long. Ty cuts it so it is $4' 8\frac{1}{2}''$ long. How much did he cut off?

10. How many inches are in 2.5 yd?

A. 30 in **B.** 90 in **C.** 7.5 in **D.** 0.07 in

11. Numerical Response A roll of duct tape holds 33 yd of tape. The length of the tape in feet is _____ ft.

C. Stretch

12. A cabinet maker cuts an 8 ft board into pieces that are each $23\frac{3}{4}''$ long. Each saw cut turns $\frac{1}{8}''$ of wood into sawdust. Can she get 4 pieces from one board? How much is left over?

13. On a staircase, each step rises $7\frac{1}{4}''$.

(a) A staircase has 13 rises. How high does it climb, in feet and inches?

(b) Another staircase must climb $8' 9''$, and each rise can be at most $7\frac{1}{2}''$. What is the fewest number of rises it can have?

14. A town is building 2.5 mi of new sidewalk. Each concrete slab is 5 ft long. How many slabs long is the new sidewalk? Use unit factors.

Extra Practice 5: Converting Length Between SI and Imperial

A. Skills

1. Which length is longer? Show how you know.
(a) 3 in or 7 cm (b) 1 ft or 0.3 m (c) 5 km or 3 mi (d) 1 yd or 90 cm

2. Estimate in your head. Show your thinking.
(a) A 40 in desk, in centimetres (b) A 10 mi bike ride, in kilometres (c) A 90 cm doorway, in inches

3. Convert. Round to the nearest tenth.
(a) 30 in to cm (b) 120 cm to in (c) 15 ft to m

(d) 2.5 m to ft (e) 8 mi to km (f) 300 km to mi

4. Change to millimetres, to the nearest tenth.
(a) $\frac{1}{4}$ in (b) $2\frac{5}{8}$ in

5. (a) Change $5' 11''$ to centimetres, to the nearest centimetre. (b) Change 175 cm to feet and inches, to the nearest inch.

B. Applying

6. A plumber has an old copper pipe that is $\frac{3}{4}$ in across. The new pipe is 20 mm across. Which pipe is wider, and by how many millimetres?

7. A family builds a backyard skating rink that is 40 ft by 24 ft. Find the length and width in metres, to the nearest tenth.

8. Round to the nearest whole number.

(a) The speed limit on a highway in Arizona is 75 mph. What is that in km/h?

(b) A school zone in Lethbridge is 30 km/h. What is that in mph?

9. Jamal drives a truck from Calgary to Great Falls, Montana. The first part, from Calgary to the border at Coutts, is about 290 km. From the border to Great Falls is about 120 mi. His odometer shows km.

(a) What is the total distance in kilometres, to the nearest kilometre?

(b) The odometer read 102 644 km in Calgary. What should it read in Great Falls?

10. LED strip lights come in a 16.4 ft roll for $12.99 or a 5 m roll for $15.49. Find the price per metre of each. Which is the better buy?

11. One mile is about

A. 1.609 m **B.** 621 m **C.** 1609 m **D.** 5280 m

12. Numerical Response A roll of ribbon is 48 in long. Its length in centimetres, to the nearest centimetre, is _____ cm.

C. Stretch

13. A sprinter runs 100 m in 12.0 s. At the same speed, how long would she take to run 100 yd, to the nearest hundredth of a second?

14. A metric cabinet has an inside width of 76 cm. Liam will cut a shelf to fit, using an inch tape.

(a) Change 76 cm to inches (to three decimals), then to the nearest $\frac{1}{16}$ in.

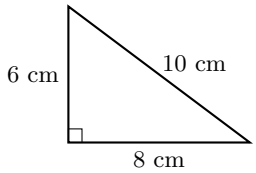
(b) Convert your answer to (a) back to centimetres. Why would $29\frac{7}{8}$ in be a safer length to cut?

15. Zoe runs at a pace of 5.5 minutes per kilometre. What is her pace in minutes per mile, to the nearest tenth? Is she faster or slower than a friend who runs a 9-minute mile?

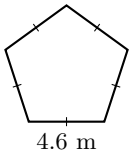
Extra Practice 6: Perimeter and Circumference

A. Skills

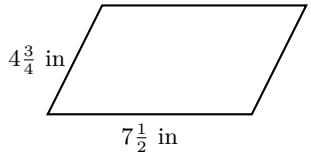
1. Find the perimeter of each shape.



(a) a right triangle



(b) a pentagon (all sides equal)



(c) a parallelogram (opposite sides are equal)

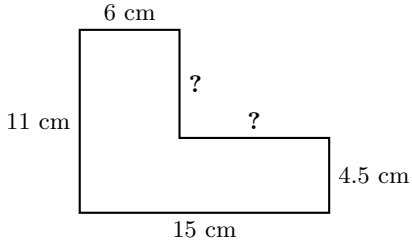
2. Find the perimeter of each rectangle or square.

(a) 25 m by 16 m

(b) a square with sides of 0.85 m

(c) 3 ft 8 in by 2 ft 7 in (answer in feet and inches)

3.



All corners are right angles.

(a) Find the two missing side lengths (marked ?).

(b) Find the perimeter. Check with a second method.

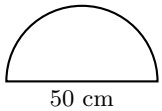
4. Find the circumference. Round to the nearest tenth.

(a) a cake pan, $d = 18$ cm

(b) a round pond, $r = 3.5$ m

(c) a pipe, $d = 1\frac{3}{4}$ in

5.



A half-circle shelf is 50 cm across its straight edge. Find the perimeter of the shelf, to the nearest tenth of a centimetre.

B. Applying

6. A soccer field is a rectangle 100 m by 60 m. In warm-up, Coach Bains has the team jog around the field 3 times. How far do they jog, in kilometres?

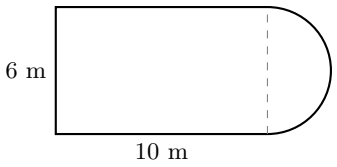
7. A rectangular deck is 4.2 m long and 3.6 m wide. One 4.2 m side is against the house. Railing goes along the other three sides and costs $45.80 per metre. How much will the railing cost before tax?

8. A wheelbarrow wheel has a diameter of 40 cm. About how many turns does the wheel make when the wheelbarrow is pushed 50 m?

9. Crown moulding goes around the top of the walls of a room 12 ft by 10 ft 6 in. It is sold in 12 ft pieces for $21.96 each. How many pieces are needed, and what is the cost?

10. Ribbon trim is sewn around the edge of a round tablecloth with a diameter of 1.5 m. The store sells the ribbon only in whole metres, at $3.25 per metre. How much will the ribbon cost?

11.



A patio is a 10 m by 6 m rectangle with a half circle on one 6 m end. The perimeter of the patio, to the nearest tenth, is

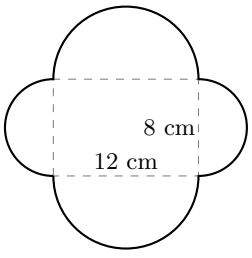
A. 30.7 m
B. 35.4 m
C. 41.4 m
D. 44.8 m

12. Numerical Response A round rug has a radius of 1.1 m. The length of fringe needed to go around its edge, to the nearest tenth of a metre, is _____ m.

C. Stretch

13. Eli can make a square garden with sides of 9 m, or a circular garden with a diameter of 11 m. Which garden needs more edging, and by how much? Round to the nearest tenth of a metre.

14.



A coaster is made from a 12 cm by 8 cm rectangle (dashed) with a half circle on each of its four sides. Find the perimeter of the coaster, to the nearest tenth. (Hint: which half circles make whole circles?)

15. Paul's backyard is a rectangle 12 m by 9.5 m. He wants a fence all the way around except for a 1 m gate (the gate comes free with the fence). Fencing costs \$14.50 per metre, and Paul has \$500.

(a) Does Paul have enough money? If not, how much more does he need?

(b) How many whole metres of fencing can Paul buy with \$500?

Extra Practice 7: Rearranging Formulas: Finding a Missing Side

A. Skills

1. Substitute, then undo to find the missing value.

(a) $P = 4s$; $P = 13$ ft. Find s.

(b) $A = lw$; $A = 84$ cm^2, $w = 7$ cm. Find l.

(c) $C = \pi d$; $C = 50$ in. Find d to the nearest tenth.

(d) $P = a + b + c$; $P = 40$ m, $a = 12$ m, $b = 15$ m. Find c.

2. Rearrange each formula to find the letter shown.

(a) $A = lw$, so $w =$ _____

(b) $P = a + b + c$, so $c =$ _____

(c) $C = 2\pi r$, so $r =$ _____

(d) $P = 2l + 2w$, so $l =$ _____

3. Find the missing side of each rectangle.

(a) $P = 30$ cm, $l = 9$ cm. Find w.

(b) $P = 17.4$ m, $w = 3.2$ m. Find l.

(c) $P = 46\frac{1}{2}$ in, $l = 12\frac{1}{4}$ in. Find w.

4. Complete the table for each circle. Round to the nearest tenth.

	Circumference	**Diameter** ($C \div \pi$)	**Radius** ($d \div 2$)
(a)	62.8 cm		
(b)	4.4 m		

5. Substitute back to check each answer. Write *correct* or *not correct*. If it is not correct, find the right answer.

(a) A rectangle has $P = 44$ m and $l = 13$ m. Lee says $w = 9$ m.

(b) A circle has $C = 40$ cm. Jo says $d = 12$ cm.

B. Applying

6. A round trampoline has a safety pad all the way around its edge. The pad is 11.3 m long. Find the diameter of the trampoline, to the nearest tenth of a metre.

7. The boards around a rectangular hockey rink have a total length of 570 ft. The rink is 200 ft long. How wide is it?

8. A carpenter used 167 in of trim all the way around a rectangular window. The window is $47\frac{1}{2}$ in wide. How tall is it?

9. A rectangular rug covers 4.32 m^2 of floor. It is 2.4 m long.

(a) How wide is the rug?

(b) Will it fit in a hallway that is 2 m wide?

10. Kaya is putting LED strip lights all the way around the railing of a rectangular deck. The lights cost \$4.50 per foot, and she spends \$162.00. The deck is 10 ft long. How wide is it?

11. Ravi wants the radius of a circle with a circumference of 30 cm. Which calculator keys are correct?

A. 30 [÷] 2 [×] [π] [ENTER]

B. 30 [÷] [(] 2 [×] [π] [)] [ENTER]

C. 30 [×] 2 [÷] [π] [ENTER]

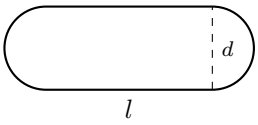
D. 2 [×] [π] [÷] 30 [ENTER]

12. Numerical Response A rectangular bulletin board has 5.2 m of border trim all the way around it. It is 1.15 m tall. Its width, to the nearest hundredth of a metre, is _____ m.

C. Stretch

13. A school track is two straight sides joined by two half circles. Its perimeter is $P = 2l + \pi d$, where l is one straight side and d is the diameter of the half circles.

(a) Rearrange the formula to find l.



(b) The track is 400 m around and $d = 64$ m. How long is each straight side, to the nearest tenth of a metre?

14. Thandi has 27 m of fencing for a rectangular garden. She wants the length to be twice the width.

(a) Explain why the perimeter is $6w$. (Hint: replace l with $2w$ in $P = 2l + 2w$.)

(b) Find the width and the length. Check your answer.

15. An arborist measures the circumference of the same poplar tree every few years. In 2022 it was 94 cm. In 2026 it was 113 cm. By about how much did the tree's diameter grow? Round to the nearest tenth of a centimetre.

Extra Practice 8: Area of 2-D Shapes

A. Skills

1. Does each job need the **area** or the **perimeter**?

(a) putting a fence around a dog run _____ (b) painting a ceiling _____

(c) laying carpet in a hallway _____ (d) gluing trim around a window _____

2. Find the area of each shape.

(a) rectangle: 14 cm by 3.5 cm

(b) parallelogram: base 8 ft, height $4\frac{1}{2}$ ft

(c) triangle: base 12 m, height 7.5 m

(d) trapezoid: parallel sides 3 in and 7 in, height $2\frac{1}{2}$ in

3. Find the area of each circle, to the nearest tenth.

(a) radius 6 cm

(b) diameter 3 ft

4. Change each area. Round to the nearest tenth if needed.

(a) 3 yd^2 to ft^2

(b) 720 in^2 to ft^2

(c) 0.6 m^2 to cm^2

(d) 50 ft^2 to m^2 (1 $m^2 \approx$ 10.76 ft^2)

5. Jen found the area of a parallelogram with a base of 10 cm, a slanted side of 6 cm and a height of 5 cm. She wrote $A = 10 \times 6 = 60$ cm^2. Find her mistake and correct it.

B. Applying

6. A lawn sprinkler turns in a full circle and sprays water 7.5 m from where it sits. What area of lawn does it water, to the nearest square metre?

7. A triangular yield sign has a base of 90 cm and a height of 78 cm.

(a) Find its area in cm^2.

(b) Change the area to m^2.

8. Ben is staining a rectangular deck that is 16 ft by 12 ft. He wants two coats. One can of stain covers 250 ft^2 and costs \$49.99.

(a) What area will the stain cover in all, for two coats?

(b) How many cans must he buy?

(c) What do the cans cost with 5% GST?

9. A flower bed is a trapezoid with parallel sides of 4.5 m and 2.5 m that are 2 m apart. One bag of bark mulch covers 0.75 m^2 and costs \$6.97. How many bags are needed, and what do they cost before tax?

10. Which shape has the largest area?

A. a square with sides of 5 m

B. a rectangle 8 m by 3 m

C. a circle with a radius of 3 m

D. a triangle with a base of 10 m and a height of 5 m

11. Numerical Response The area of the top of a 14 in pizza (its diameter is 14 in), to the nearest square inch, is _____ in^2.

C. Stretch

12. Marco says, "If you double the side length of a square, you double its area." Test his idea with a square that has sides of 3 ft. Is he right? Explain.

13. A room is 4.2 m long and 3.5 m wide. The flooring Lena likes is sold by the square foot, at \$3.29 per ft^2.

(a) Find the area in m^2, then change it to ft^2 using 1 $m^2 \approx 10.76$ ft^2. Round to the nearest tenth.

(b) Find the cost of the flooring before tax.

14. A rectangular banner for a school event must have an area of 3 m^2. It is 1.2 m high.

(a) How long must the banner be? Check your answer.

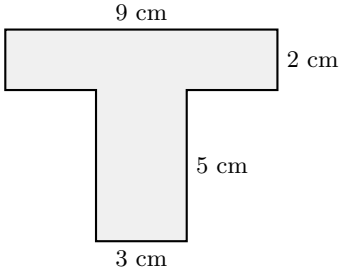
(b) Fabric costs \$12.40 per m^2. The sign maker buys 10% extra for hems. What does the fabric cost?

Extra Practice 9: Composite and Irregular Shapes

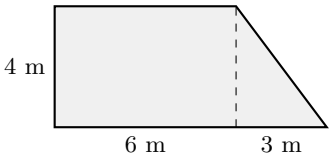
A. Skills

1. Find the area of each composite shape. All the corners that look square are square.

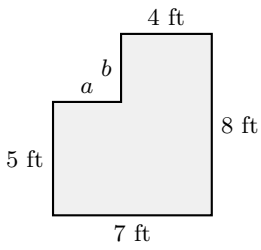
(a)



(b)



2. The plan of a storage room is shown. All the corners are square.



(a) Find a and b.

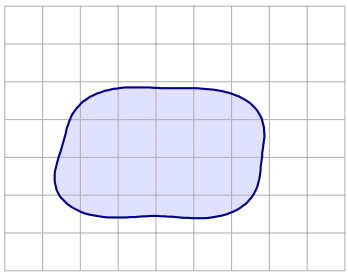
(b) Find the area of the room.

(c) Check with whole minus cut-out.

3. A rug is a rectangle 2 m long and 1.5 m wide, with a semicircle on one 1.5 m end. Find the area of the rug, to the nearest hundredth of a square metre.

4. A metal plate is a rectangle 20 cm by 12 cm. A round hole 6 cm across is drilled through it. Find the area of the metal that is left, to the nearest hundredth of a square centimetre.

5. Nadia traced a leaf on centimetre grid paper. Each square is 1 cm^2.

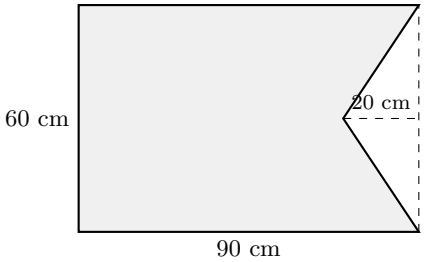


(a) Estimate the area of the leaf.

(b) Sam says the leaf is about 24 cm^2. What mistake did he make?

B. Applying

6. A swallowtail banner is cut from a rectangle of fabric 90 cm long and 60 cm wide. A triangle is cut out of one end. The triangle has a base of 60 cm and a height of 20 cm.



(a) Find the area of the banner in cm^2.

(b) Change the area to m^2. (1 m^2 = 10 000 cm^2)

7. A kitchen floor is a rectangle 13 ft 6 in by 10 ft, plus a pantry 4 ft by 3 ft 6 in. Vinyl plank flooring comes in boxes that cover 24 ft^2 each, for $56.99 a box.

(a) Find the area of the floor.

(b) How many boxes are needed, and what do they cost (before tax)?

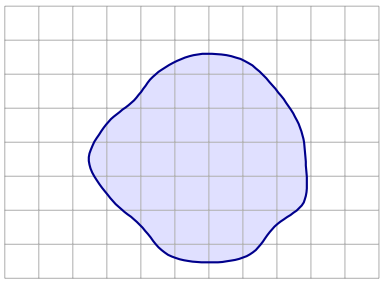
8. A rectangular pool is 8 m by 4 m. A deck 1.5 m wide goes all the way around it.

(a) What are the length and width of the outside edge of the deck?

(b) Find the area of the deck.

(c) One can of deck stain covers 20 m^2. How many cans are needed for one coat?

9. A park map shows a lake. Each square stands for 20 m by 20 m.



(a) What area does one square stand for?

(b) Estimate the area of the lake.

(c) Draw the smallest rectangle around the lake. Is your estimate reasonable?

10. A wall is 10 ft long and 8 ft high. It has a door 3 ft wide and 7 ft high. The area of the wall to be painted is

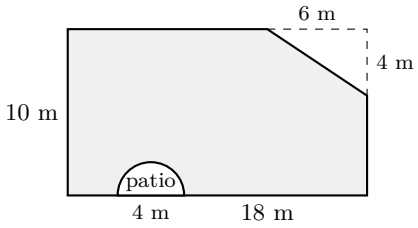
A. 70 ft^2 **B.** 59 ft^2 **C.** 101 ft^2 **D.** 26 ft^2

11. Numerical Response An arched doorway is a rectangle 1.2 m wide and 2.4 m high, with a semicircle on top. The area of the doorway, to the nearest hundredth of a square metre, is _____ m^2.

C. Stretch

12. A rectangular garden is 12 m long. A shed 3 m by 3 m sits in one corner of it. The area of the garden **not** covered by the shed is 87 m^2. How wide is the garden?

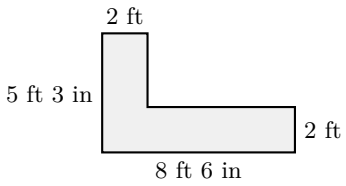
13. A lawn is a rectangle 18 m by 10 m. A triangle with legs 6 m and 4 m is cut off one corner for a garden, and a semicircular patio 4 m across sits along one side.



(a) Find the area of the lawn, to the nearest hundredth of a square metre.

(b) One bag of fertilizer covers 50 m^2. How many bags are needed?

14. An L-shaped countertop is 2 ft deep. Its two outside edges are 8 ft 6 in and 5 ft 3 in long, as shown.



(a) Find the area of the countertop in ft^2. Be careful not to count the corner twice.

(b) Check your answer by splitting the other way.

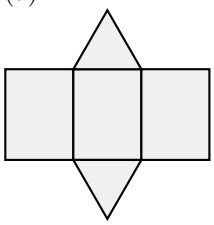
(c) Quartz costs \$75.00 per ft^2, installed. What will the countertop cost?

Extra Practice 10: Nets and Surface Area of Prisms

A. Skills

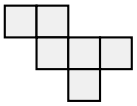
1. Name the 3-D object that each net folds into, and give its number of faces.

(a)



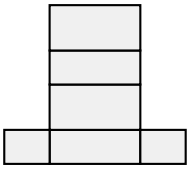
_____

(b)



_____

(c)



_____

2. Find the surface area of each cube.

(a) edges 7 cm

(b) edges 1.5 m

(c) edges $5\frac{1}{4}$ in

3. Find the surface area of each rectangular prism.

(a) 12 cm by 9 cm by 5 cm

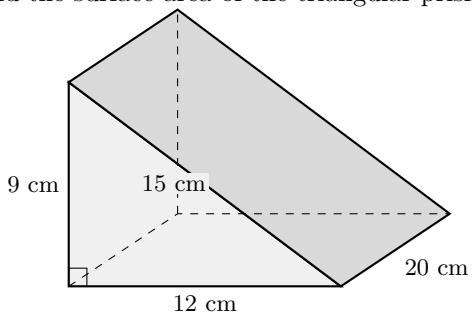
(b) 3 m by 2 m by 0.5 m

(c) 1 ft 6 in by 1 ft by 8 in (answer in in^2)

4. A storage tote is 50 cm long, 40 cm wide and 30 cm high, with a lid. Complete the face table, then write the surface area in m^2.

Faces	**Size**	**Area of one**	**Area of two**
top & bottom			
front & back			
two ends			
		Surface area	

5. Find the surface area of the triangular prism.



B. Applying

6. A metal baking pan has no top. It is 14 in long, 10 in wide and 2 in deep. How much metal is in the pan?

7. Nia is painting the walls and ceiling of a storage room with one coat. The room is 3 m long, 2.4 m wide and 2.4 m high. The door is 0.8 m by 2 m and is not painted. A 1 L can of paint covers 10 m^2 and costs \$18.99.

(a) Find the area to be painted.

(b) How many cans must she buy, and what do they cost before tax?

8. A pup tent is a triangular prism. The front triangle is 6 ft wide at the bottom and 4 ft high. Each sloping side is 5 ft, and the tent is 7 ft long.

(a) Find the area of fabric for the whole tent, including the floor.

(b) Change your answer to square yards, to the nearest tenth.

9. A shipping crate with a lid is built from plywood. It is 4 ft long, 3 ft wide and $2\frac{1}{2}$ ft high. Plywood comes in 4 ft by 8 ft sheets that cost \$42.97 each.

(a) Find the surface area of the crate.

(b) How many sheets are needed (by area), and what do they cost before tax?

10. A box is 10 cm long, 10 cm wide and 4 cm high. What is its surface area?

A. 400 cm^2 **B.** 180 cm^2 **C.** 360 cm^2 **D.** 280 cm^2

11. Numerical Response A small garden shed is shaped like a cube with edges 2.4 m long. The surface area of the whole shed (all 6 faces), to the nearest tenth of a square metre, is _____ m^2.

C. Stretch

12. Two boxes hold the same amount. Box A is 20 cm by 10 cm by 5 cm. Box B is a cube with edges 10 cm.

(a) Find the surface area of each box.

(b) Which box uses less cardboard? What percent less, to the nearest tenth of a percent?

13. A closed box has a base 6 cm long and 4 cm wide. Its surface area is 148 cm^2. How high is the box? Check your answer.

14. A plastic bin is 60 cm long, 40 cm wide and 30 cm high, with a lid. Find its surface area in cm^2, in m^2, and in ft^2 (use 1 $m^2 \approx 10.76$ ft^2; round to the nearest tenth).

Extra Practice 11: Surface Area of Cylinders, Pyramids and Cones

A. Skills

1. Write the formula for the area of each piece.

Piece	Area formula
(a) one circle (the end of a cylinder or the base of a cone)	
(b) the curved side of a cylinder	
(c) one triangle face of a square pyramid	
(d) the curved side of a cone	

2. Find the surface area of each closed cylinder, to the nearest tenth.

(a) radius 6 cm, height 15 cm

(b) a hockey puck: diameter 3 in, height 1 in

3. Find the surface area of each square pyramid.

(a) base edge 10 cm, slant height 12 cm

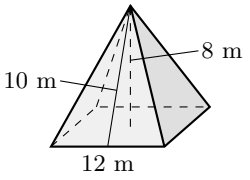
(b) base edge 4 ft, slant height $5\frac{1}{2}$ ft

4. Find the surface area of each closed cone, to the nearest tenth.

(a) radius 3 m, slant height 5 m

(b) diameter 8 in, slant height 9 in

5. Find the surface area of the square pyramid. Which length in the picture do you *not* need?



B. Applying

6. A round hay bale is a cylinder with a diameter of 1.5 m and a width (height) of 1.2 m. Net wrap covers only the curved side, not the two flat ends.

(a) How much wrap covers one bale, to the nearest tenth?

(b) How much wrap covers 40 bales, to the nearest square metre?

7. A traffic cone (without its square base) is a cone with no bottom. It is $10\frac{1}{2}$ in across at the bottom and has a slant height of $27\frac{1}{2}$ in. How much orange plastic is in the curved side, to the nearest square inch?

8. A round above-ground pool has a diameter of 15 ft and walls 4 ft high. The vinyl liner covers the floor and the inside of the walls (no top). How much vinyl is in the liner, to the nearest tenth?

9. A children's play tent is a cone with no floor. It is 1.6 m across the bottom and has a slant height of 2 m.

(a) How much canvas is in the tent, to the nearest tenth?

(b) Canvas costs \$8.50 per m^2. Use your answer to (a) to find the cost of the canvas.

10. Each fence post on a ranch has a small wooden cap shaped like a square pyramid. The base edge is 6 in and the slant height is 4 in. Only the 4 triangles get painted.

(a) Find the area to paint on one cap.

(b) Find the area for 24 caps, in in^2 and in ft^2.

11. A section of drainpipe has no ends. Which formula gives the area of metal in it?

A. $2\pi r^2 + 2\pi rh$ **B.** $2\pi rh$ **C.** $\pi r^2 + 2\pi rh$ **D.** πrs

12. Numerical Response A closed cone has a radius of 5 cm and a slant height of 9 cm. Its surface area, to the nearest square centimetre, is _____ cm^2.

C. Stretch

13. A candle maker wraps each candle in foil, covering every face. A pyramid candle has a square base 6 cm on each side and a slant height of 8 cm. A cylinder candle has a diameter of 6 cm and a height of 8 cm.

(a) Which candle needs less foil?

(b) How much foil is saved by wrapping 200 pyramid candles instead of 200 cylinder candles? Answer in m^2, to the nearest tenth.

14. Kai says, "If you double both the radius and the height of a closed cylinder, its surface area doubles." Test his idea with $r = 2$ cm and $h = 5$ cm. Is he right?

15. A square pyramid has a surface area of 96 cm^2. Its base edge is 6 cm. Find its slant height. Check your answer.

Extra Practice 12: Unit 3 Review

Practice Test 2. Treat this like the unit test. Show your work. Use the conversion numbers from the review summary and the $\boxed{\pi}$ key.

A. Skills

1. Fill in the blanks.

(a) 5.08 m = _____ cm

(b) 7500 m = _____ km

(c) 36 mm = _____ cm

(d) 7 ft = _____ in

(e) 4 yd = _____ ft

(f) 87 in = _____ ft _____ in

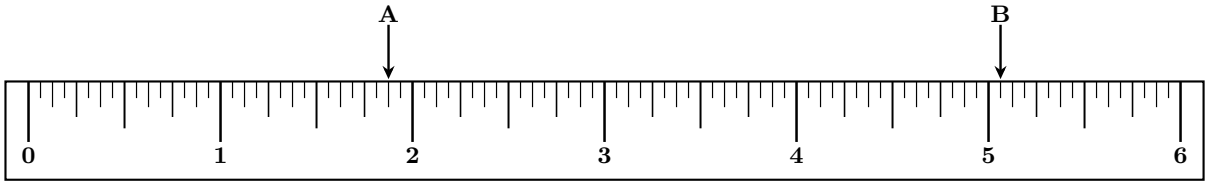
2. Find each answer in simplest form.

(a) $3\frac{7}{8} + 2\frac{1}{2}$

(b) $5\frac{1}{16} - 2\frac{3}{8}$

(c) $4 \times 1\frac{3}{4}$

3. The ruler is real size.



(a) Read the length at each arrow. A _____ B _____

(b) How far apart are the arrows?

4. Convert. Round to the nearest tenth unless told otherwise.

(a) 7 in to centimetres

(b) 1.5 mi to kilometres

(c) 3 m to feet (1 m $\approx$ 3.28 ft)

(d) $6' 2''$ to centimetres (nearest centimetre)

5. Find each area.

(a) A parallelogram with a base of 7.5 cm and a height of 4 cm.

(b) A triangle with a base of $5\frac{1}{2}$ ft and a height of 4 ft.

(c) A circle with a radius of 6 in (to the nearest tenth).

(d) A trapezoid with parallel sides of 2.5 m and 4.5 m and a height of 2 m.

B. Applying

6. Mia is $5'\,2''$ tall and her brother is $6'\,1''$ tall.

(a) How much taller is her brother, in inches?

(b) What is that difference in centimetres, to the nearest tenth?

7. A round flower bed is 3.2 m across. Plastic edging is sold in 1.5 m pieces for $4.49 each. How many pieces are needed to go around the bed, and what do they cost before tax?

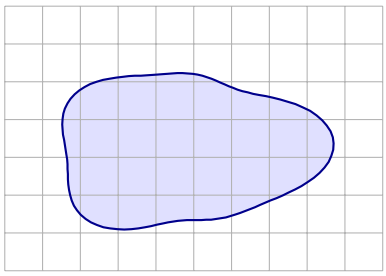
8. A rectangular outdoor rink uses 180 ft of boards around the outside. The rink is 35 ft wide. How long is it? Check your answer.

9. A rectangular lawn is 15 m by 9 m. A round pond 3 m across sits in the lawn.

(a) Find the area of grass, to the nearest hundredth.

(b) One roll of sod covers 0.75 m^2. How many rolls are needed?

10. A plan for a park shows a new skating pond on a grid. Each square is 2 m by 2 m.



(a) What area does one square stand for?

(b) Estimate the area of the pond.

11. A closed shipping box is 24 in long, 18 in wide and 12 in tall.

(a) Find its surface area in square inches.

(b) Change your answer to square feet.

12. A can has a radius of 4 cm and a height of 10 cm. The area of the label around its side is about

A. 251.3 cm^2 **B.** 125.7 cm^2 **C.** 502.7 cm^2 **D.** 351.9 cm^2

13. Numerical Response A square pyramid has a base 6 cm on each side and a slant height of 9 cm. Its total surface area is _____ cm^2.

C. Stretch

14. Kiran is painting the walls of a bedroom. The floor is 12 ft by 10 ft and the walls are 8 ft high. The room has a door $3'\,0''$ by $6'\,8''$ and a window $4'\,0''$ by $3'\,0''$ that will not be painted.

(a) Find the area of the four walls before the door and window are taken out. (Hint: the walls unroll into one long rectangle as long as the perimeter of the floor.)

(b) Find the area to be painted.

(c) One gallon of paint covers 350 ft^2. Kiran wants 2 coats. How many gallons must he buy? At \$54.99 a gallon, what is the cost before tax?

15. A round steel storage tank at a water plant is a cylinder 6 m across and 12 m tall, with a cone roof. The roof has a slant height of 4 m and the same radius as the tank.

(a) Find the area of the side of the tank, to the nearest hundredth.

(b) Find the area of the roof.

(c) Steel sheeting costs $28.50 per square metre. Find the cost of the steel for the side and roof (the floor is concrete).